Question	Answer	Marks	Partial Marks
10(a)	1600	**3**	**B2** for answer figs 16 or **M2** for 90.72 ÷ (figs45 × figs3 × figs42) or **M1** for volume = figs 45 × figs 3 × figs 42 isw
10(b)	62.8 or 62.83 to 62.84	**3**	**M2** for $\frac{\pi \times 10^2 \times 30}{15000} \times 100$ or **M1** for $\pi \times 10^2 \times 30$
10(c)	12.9[0]	**3**	**B2** for 86 OR **M2** for $\frac{98.9}{1+\frac{15}{100}} \times 0.15$ oe or $98.9 - \frac{98.9}{1+\frac{15}{100}}$ oe or **M1** for $\left(1+\frac{15}{100}\right)a = 98.9$ oe isw
10(d)	50	**2**	**M1** for 3540 ÷ 70.8
11(a)	$\frac{48}{x}$ final answer	**1**	Accept 48 ÷ x
11(b)	$their(a) - \frac{60}{x+2} = 4$ oe	**M1**	FT *their* (a) provided expression in x
	$48(x+2) - 60x = 4x(x+2)$ oe	**M2**	**FT** *their* 3 term eqn with algebraic denominators, x and $x + 2$, for M2 or M1 **M1** for common denominator $x(x+2)$ oe seen or any two terms in a 3 term equation from ± 48 (x + 2) , ± 60x , ± 4x(x + 2) oe seen
	$48x + 96 - 60x = 4x^2 + 8x$ oe leading to $x^2 + 5x - 24 = 0$	**A1**	With brackets expanded and no errors or omissions seen
11(c)	$(x-3)(x+8)$	**B2**	**B1** for $x(x + 8) - 3(x + 8)$ or $x(x - 3) + 8(x - 3)$ or $(x + a)(x + b)$ [= 0] where $ab = -24$ or $a + b = 5$ [a, b integers]
	3 and − 8	**B1**	
11(d)	12	**1**	
12(a)	17	**3**	**M2** for $3 \times 2x^2 - 7$ or better isw or **M1** for $3 \times 2x^2$ oe or $kx^2 - 7$ seen

Question	Answer	Marks	Partial Marks
12(b)(i)	13.4 or 13.41 to 13.42	**3**	**M2** for $\sqrt{(-5-7)^2+(8-2)^2}$ oe or **M1** for $(-5-7)^2+(8-2)^2$ oe
12(b)(ii)	$[y=]\ 2x+5$ final answer	**4**	**M1** for [gradient of AB =] $\frac{8-2}{-5-7}$ oe **M1dep** for gradient $p = -1 \div their - \frac{1}{2}$ oe **M1dep** on previous M1 for substituting (−1, 3) into $y = their\ px + c$ oe where $their\ p \neq 0$
12(b)(iii)	(5, 0)	**4**	**B3** for $\overrightarrow{AD} = \begin{pmatrix}-2\\-2\end{pmatrix}$ or $\overrightarrow{DA} = \begin{pmatrix}2\\2\end{pmatrix}$ or coordinates of C (−7, 6) and $\left[\overrightarrow{CD} =\right] \begin{pmatrix}12\\-6\end{pmatrix}$ oe seen or **B2** for $a = b = 2$ soi or coordinates of C (−7, 6) or **M1** for $a = b$ oe soi or for $a^2+b^2=\left(\sqrt{8}\right)^2$ oe or cos 45 = $\frac{a}{\sqrt{8}}$ oe or for $\left[\overrightarrow{DC} =\right] \begin{pmatrix}-12\\6\end{pmatrix}$ or $\left[\overrightarrow{CD} =\right] \begin{pmatrix}12\\-6\end{pmatrix}$ seen or $\frac{y-8}{x--5}=1$ oe or $\frac{y-2}{x-7}=1$